ODONT 541.1

CARIOLOGY

Nutrition and Caries

Lecture #9

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Office Hours: **By appointment**

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Ecologic Plaque Hypothesis

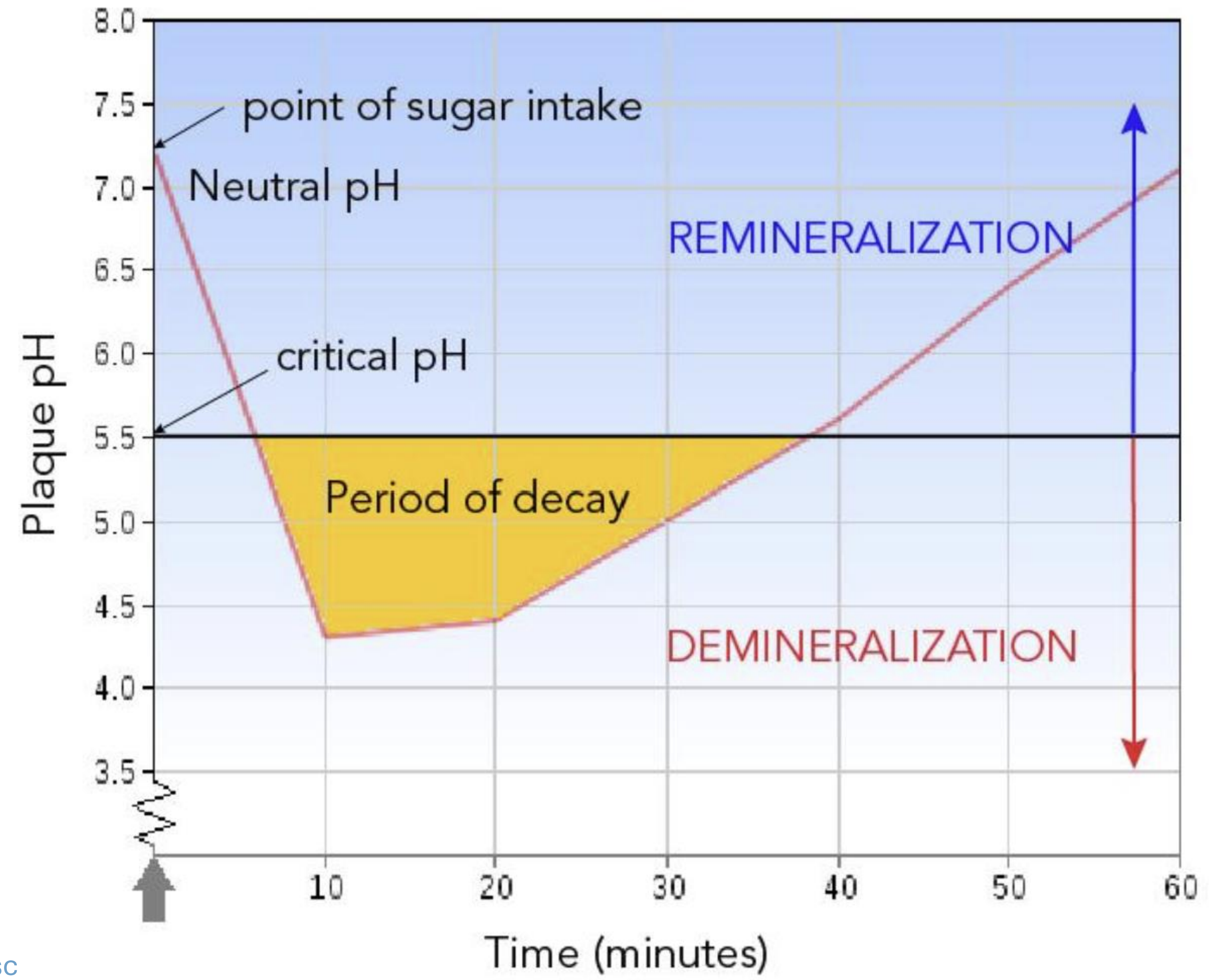
- DIET is what causes the ecologic shift.
  - **Frequent** exposure to fermentable carbs results in microbial succession in the plaque
  - The same person, with a different diet, will have a different microbial profile in their plaque

Caries is best understood as **a diet-driven ecological shift in the plaque biofilm**, rather than a classic infectious disease caused by a single organism.

The Stephan curve (1940S) was based on:

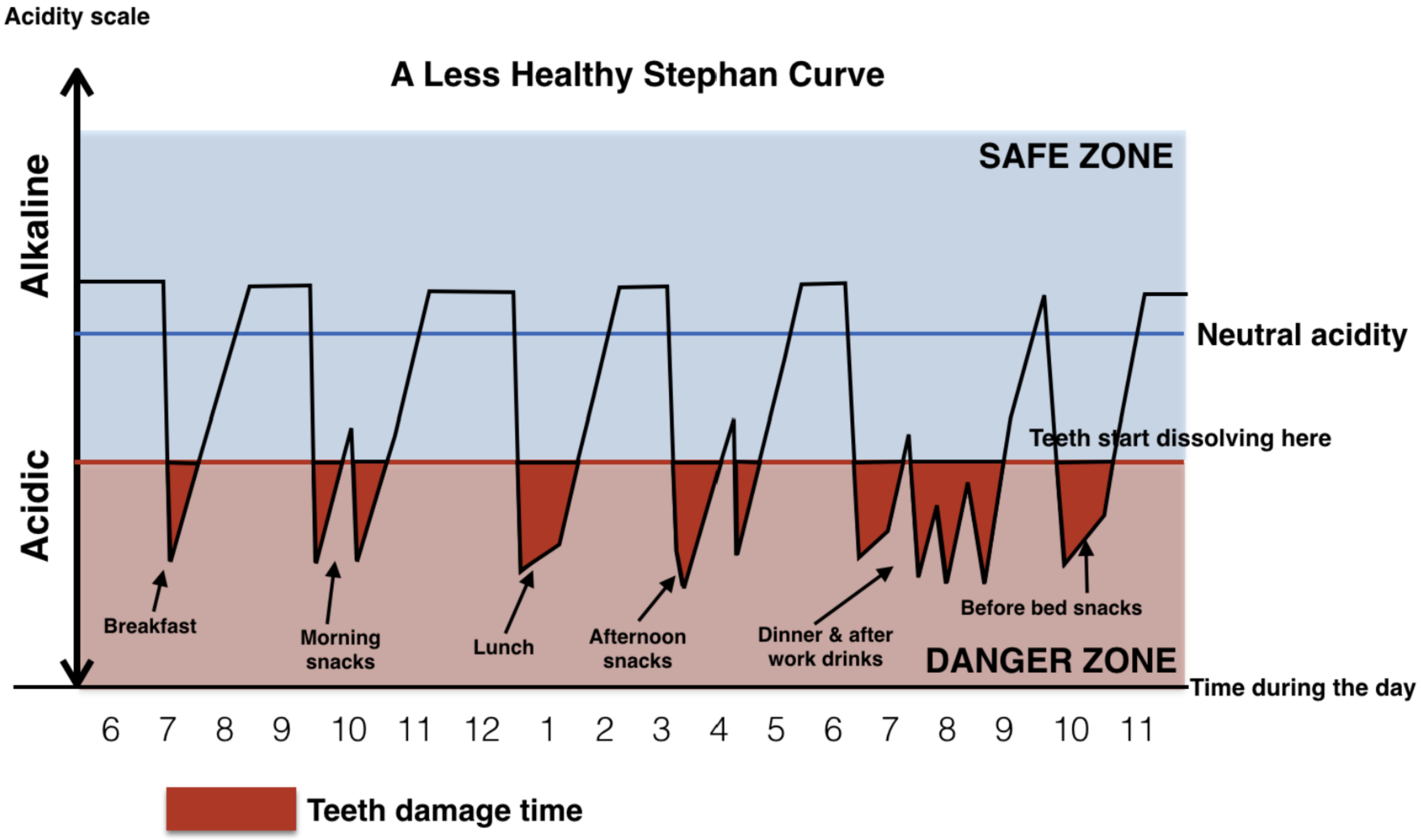
- Exposure to liquid **glucose** (not sucrose) because it is experimentally simple. It's a monosaccharide that diffuses into plaque quickly, is rapidly metabolized, and therefore causes a fast drop.
- Explains "What happens when bacteria are given a simple fermentable carbohydrate"
- Does not fully explain "what happens when a person eats candy"

Sucrose would result in a delayed dip but a deeper and longer curve because its promotes EPS formation and microbial succession.



Individuals have different curves depending on:

- Salivary state
    - **Xerostomic patients experience 2-3x slower pH recovery**
- Types of dietary exposure
- Their critical pH threshold (root exposure, FA conversion)



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Vipeholm Studies

- 1938: National dental service of Sweden was established. Caries were widespread. There was a general suspicion that high sugar diets caused caries but it was not yet proven.
- 1945: The medical board commissioned a study to determine if sugar causes caries
- 1940s-50s: Human experiments on intellectually disabled residents of the Vipeholm hospital in Sweden ensued.
  - Residents were given "extra sticky toffee" that was specially made for the study to be extra sticky.
  - Experiments were sponsored by sugar industry and dental community to determine relationship between carbs and caries

Vipeholm Studies

- Outcomes:
    - **Frequency of sugar intake matters more than the total amount**
    - **Sticky sugars between meals significantly increase the amount of caries**
    - **Sugars with meals are much less harmful**
- The sugar industry was not happy with the outcomes. Decades later, ethical concerns with the study design were raised.
- Since the Vipeholm studies, Lördagsgodis ("Saturday sugar") became recommended in Sweden.
    - Avoid candy during the week. Visit the candy store on Saturdays.
    - Temporarily decreased consumption but currently, Sweden has the highest candy consumption in the world.

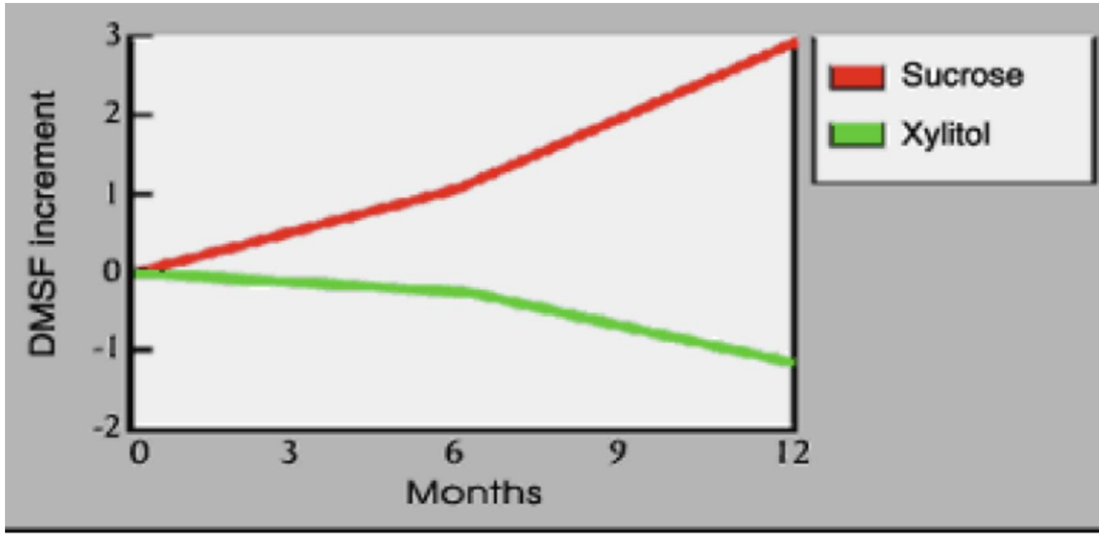
Hopewood House Studies

- Australia in the 1940s-60s.
- The Hopewood House was a home in rural South Whales, Australia that provided care for economically deprived children from birth through 12 years of age
- While living in Hopewood House, the children consumed a lacto-ovo vegetarian diet with emphasis on whole grains and raw vegetables. Sugars and refined starches were severely limited
- Researchers compared:
  - Caries rate of Hopewood House children with children in public schools without dietary restrictions
  - Caries rate of Hopewood House children while they were in residence and after they left the program.
- Findings:
  - Children in Hopewood House had a very low caries rate, even without fluoride treatment
  - However, when they later adopted a conventional diet, caries rate rapidly increased
- Showed that a strict diet alone can suppress caries. But that control is fragile
- Confirmed that dietary sugar is a primary driver of dental caries

Turku Studies (1970s)

- Finland in the 1970s
- Participants were assigned diets (and eventually chewing gums) where sweeteners were either sucrose, fructose, or xylitol to compare their individual cariogenic effects
  - Sucrose = highest caries incidence
  - Fructose = moderate
  - Xylitol = very low

- Showed that not all sugars are equally cariogenic and that xylitol is not only non cariogenic, it is actually cariostatic.
- Was a foundational study leading to increase in products with sugar substitutes, specially xylitol containing products.



Ref: Scheinin, A., Makinen., K.K., Tammisalo, E., & Maari, R. Acta Ondontol. Scand. 1975; 33: 269

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Kashket & DePaola Cheese Studies

- After inducing a rapid drop in pH, test subjects were given cheese and controls were not.
  - Cheese group had a rapid rise in pH within minutes
- Mechanisms
  - Salivary stimulation from HARD cheeses clears sugar and increases buffering capacity
  - Calcium and phosphate delivery increases saturation in saliva
  - Casein helps stabilize calcium/phosphate in plaque and maintain supersaturation near the enamel
  - Contains arginine (see later)

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WHO Review in the 2010s-2020s

- Systematic review by WHO, from which they concluded that:
  - Dental caries is the most common non communicable disease worldwide
  - Even though caries is preventable, it poses a major healthy burden in most countries and populations
  - The consumption of free sugars is the most common risk factor for dental caries
    - It is also a risk factor for other non communicable diseases, such as: DM2, cardiovascular disease, obesity, non alcoholic liver disease, etc.
      - Therefore, reducing free sugar consumption
- Therefore, WHO recommends:
  - Children under 2 should not consume ANY sugar sweetened beverages
  - Implementation of financial dietary incentives to reduce free sugar consumption
    - 100+ countries now have some form of taxation on sugar sweetened beverages, affecting about 60% of the world population
      - A few US cities (Philadephia, Berkely) have similar policies
    - Reformulation incentives for manufacturers to change their products ingredients
    - Labeling and marketing restrictions for public education
  - **Limit free sugar intake to no more than 5-10% of total caloric intake**

Intrinsic Acids- GERD

- Intraorally, erosion pattern often presents as
  - Cupping of posterior occlusal cusps
  - Palatal surfaces of maxillary anteriors
- Silent GERD
- Gastric pH is 1-2
  - Frequency vs quantity
- How to measure control
- What does acid erosion look like on teeth
  - Active vs non active
- Often occurs at night
  - Longer exposure and less buffered due to positioning and lower salivary flow

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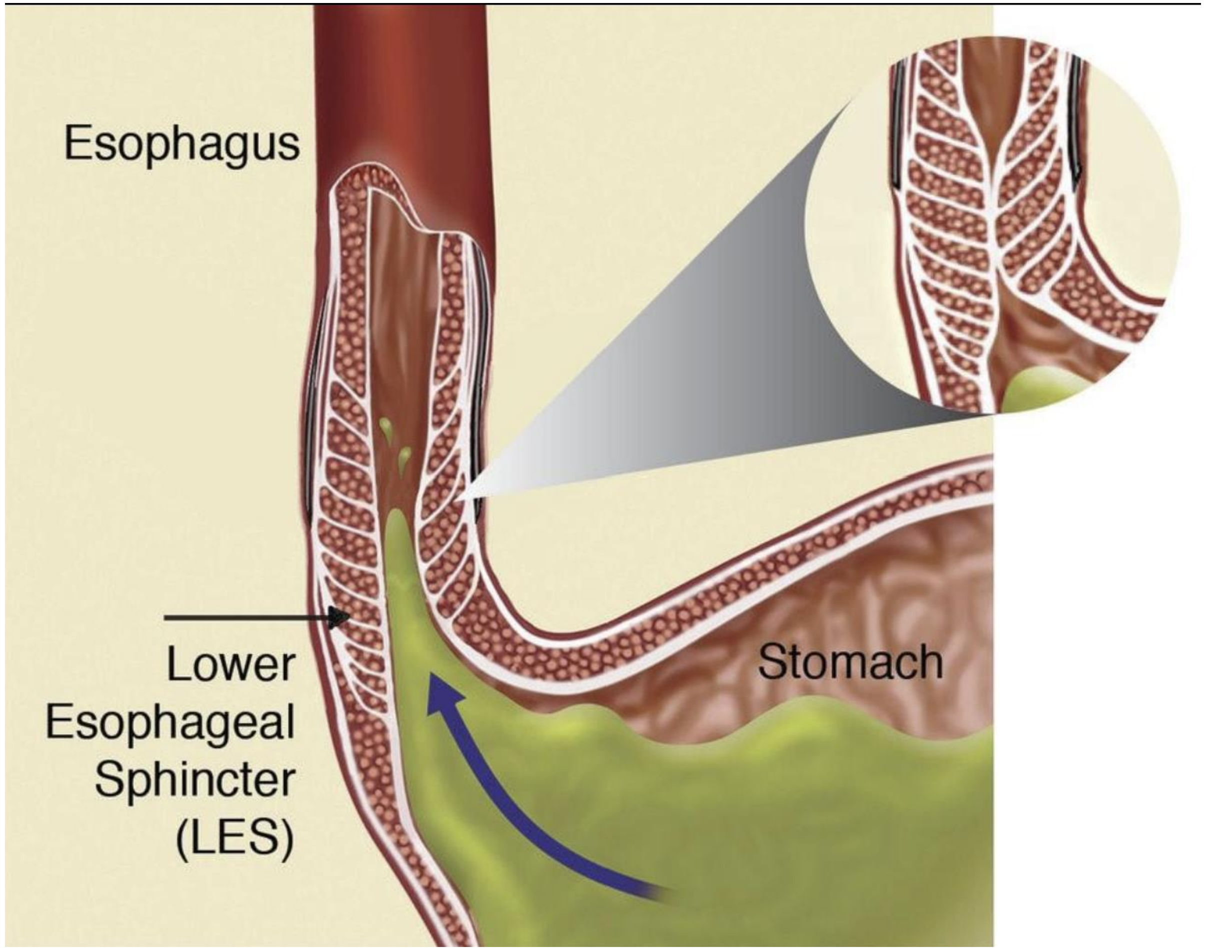
Intrinsic Acids- GERD

- Things that make GERD worse
  - Decreasing Lower Esophageal Sphincter tone
    - Chocolate
    - Alcohol
    - Caffeine
  - Delay gastric emptying
    - High fat meals
  - Increase gastric acidity/irritation
    - Spicy foods
    - Carbonated beverages



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Intrinsic Acids- GERD

- Some patient have "silent reflux" (laryngopharyngeal reflux or LPR) without heartburn
  - LPR often occurs during the daytime while upright. Multiple short exposures (compared to nighttime GERD)
  - The dentist may be the first to notice intraoral signs of reflux, warranting referral to PCP with reference to any professionally documented signs or symptoms
  - Other common signs include chronic throat cleaning and mucus, chronic cough, "lump in throat" sensation, post nasal drip.

- Active/recent acid exposure will result in an eroded tooth surface that is shiny and smooth like glass.
  - If inactive, the surface will develop a more matte appearance over time

Intrinsic Acids- Bulimia nervosa, Hyperemisis

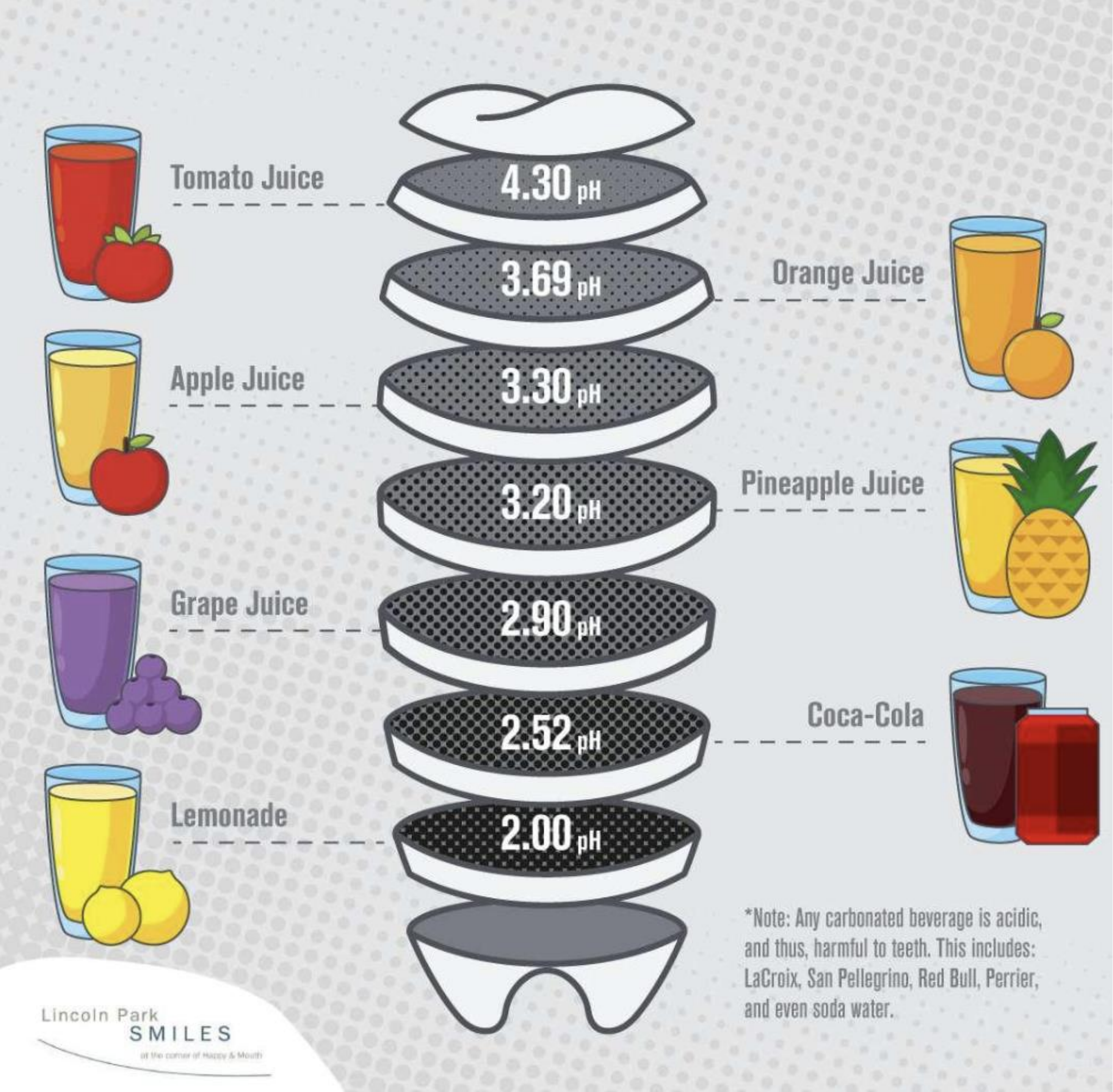
- Similar presentation to GERD but varying in severity depending on duration of disease.
  - Bulimia has additional factors of reduced salivary flow from dehydration and frequent carbohydrate intake during binge episodes.
- Hyperemisis
- **No brushing for 30 minutes after acid exposures to avoid abrasion of softened/eroded enamel.**
- Acid erosion around existing restorations can result in "proud" restorations.
  - Amalgams look like islands. Tooth structure around the restoration is eroded away but there are little to no open marigns or caries.

Extrinsic Acids - SSBs

- Erosive pattern more common on the facial/labial surfaces of teeth
- Sugar sweetened beverages
  - SSB intake -> 2 to 2.5x higher caries prevalence
    - Frequency and sipping behavior
  - Soda
  - Juice
  - Flavored water
  - Sports drinks and energy drinks
    - Caffeine promotes xerostomia
- Replacing sugar sweetened beverages with water is one of the highest ROI behavioral changes
  - Supports salivary production
  - Decreases erosion risk
  - Decreases caries risk
  - Is generally good for systemic health
- Specially avoid SSBs and/or carbonated drinks 2-3 hours before bed

- Tap water: 6.5-8.5
- Fiji and Evian: 7-7.5
- Dasani and Aquafina: 5.0-5.5
- Smart water: 6.5-7.0
- LaCroix: 4.5-5.5
- Red Bull: 3.0-3.5
- Black coffee and tea: 5s
- Milk: 6.5-6.8

- Avoid constant sipping.
- Consider rinsing with water afterwards.
- Drink sweet/acidic beverages with meals.



Extrinsic Acids – Other

- Cough drops
  - Held in mouth for long periods over many days
- "Swimmer's erosion"
  - Overchlorinated pools
- Lemon sucking
  - **Citric acid does not only erode enamel. It also binds calcium.**

Fiber

- Current research ongoing about fiber to carbohydrate ratios
  - Higher fiber = lower caries risk
  - Chewing on fibrous foods
    - Increases salivary flow
    - Increases mechanical food clearance
    - Decreases fermentable substrate. People with high fiber diets tend to avoid refined carbohydrates and frequent snacking
  - Best way to kick a bad habit (sugar) it to replace it with a better habit (fiber).
    - Fiber helps reduce sugar cravings
  - Fiber at the end of meals helps with pH recovery

Infants and Children

- Children don't choose their diet. They inherit patterns from their caregivers.
- Caregivers can model good
  - Food choices
  - Eating frequency and times
  - Beverage habits
- Specific guidance:
  - Avoid juices and sodas in sippy cups/bottles throughout the day. Reserve for mealtime.
    - Encourage free access to water instead
  - No bedtime sugar exposure
  - After brushing, nothing but water

Dietary changes with tooth loss

- Tooth loss -> reduced masticatory efficiency
  - Avoidance of hard and fibrous foods
  - Shift towards softer and more processed foods
- Tooth loss -> bones loss -> gingival recession and root exposure
- Tooth loss is cumulative over time
  - Older patients are more likely to be xerostomic
- No prosthetic is as good as a real tooth
  - All dental prosthetics (dentures, bridge, implants) create additional protected sites
    - Those protected sites are often against root structure
- For patient's who temporarily require a soft food diet, recommend:
  - Eggs
  - Yogurt
  - Cheese
  - Cooked vegetables
  - Soups, stews, and blended foods without added sugars

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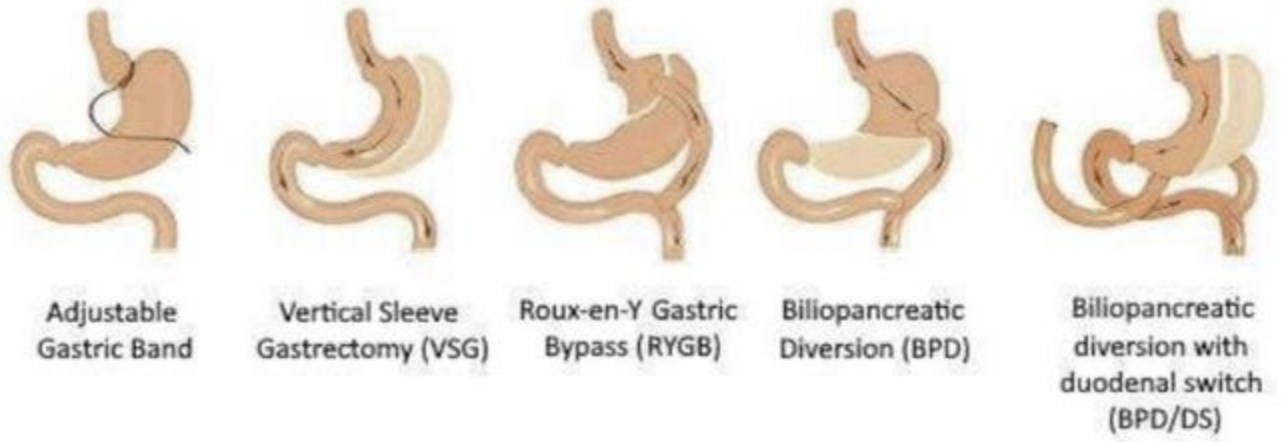
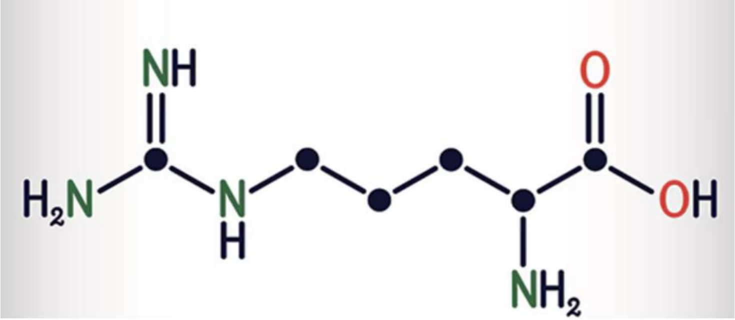
Bariatric Surgery

- After bariatric surgery, patients are instructed to eat frequent, small meals.
- Reflux and regurgitation are common
- Fluids are strongly encouraged but many choose flavored drinks instead of water
    - Reduced fluid intake (smaller gastric volume) leads to xerostomia
- Vitamin supplementation is often needed
    - Many vitamins are sugary/acidic



Arginine



- Certain oral bacteria (especially commensals like *Streptococcus sanguinis* and *Streptococcus gordonii*) metabolize arginine via the arginine deiminase system (ADS):
  - **Arginine → ornithine + ammonia + CO_2**
    - **Ammonia (NH_3) raises plaque pH**
    - This directly counteracts acid production from cariogenic bacteria like *Streptococcus mutans* and helps buffer the Stephan curve and promotes faster recovery after an acid challenge.
- Most commonly studied formats:
  - 8% arginine without fluoride
    - Insufficient evidence to support arginine alone as a caries-preventive agent.
  - 1.5% arginine + fluoride toothpaste
    - Well supported as a synergistic agent with fluoride, providing better caries prevention and better remineralization.
- Arginine is found at higher concentrations in the saliva of people who are caries free versus the low levels detected in people with a history of dental decay

Arginine

- A naturally occurring amino acid. Dietary prebiotic sources include:
  - Nuts and seeds (pumpkin, sesame, almonds, walnuts, peanuts)
    - A caries protective snack that is both low carb and high arginine
  - Meats, poultry, fish, seafood
  - Dairy
  - Legumes
  - Chicken
  - Breast milk
  - Dietary supplement commonly included in baby food and formula

- Arginine-rich foods only help if they influence **plaque metabolism timing.** It is not the volume of arginine that matters but the timing/frequency.
  - Eating nuts/cheese between meals can help buffer acids
  - Eating arginine foods with continued high sugar exposure has limited benefit
    - Frequent sugar intake still overwhelms alkalinogenic effects

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The caries rate is roughly 2% in dogs.

Dietary findings in that 2% include high frequency consumption of:

- Pup cups
- Bananas
- Marshmallows and candies
- Cereal
- Peanut butter
- Bread
- Carrots
- Apples
- Home cooked carbohydrate rich diets

Dietary Journaling

Homework:

- Keep a diet journal for at least 3 consecutive days
- We are focusing on frequency, type and duration of exposure
- For each day
    - Log each cariogenic exposure
        - What type of food?
        - Exposure time
            - How long did it take you to finish it
        - Recovery time
            - See numbers for each category
        - Total exposure time.
    - **Add up total exposure and recovery times for the day**
- **How MANY sugar attacks did you experience per day?**
- **How many GRAMS of sugar did you consume per day?**

Dietary Journaling

- Sugar sweetened beverages
    - 30 minutes
- Sticky sweets (candy, cake, bananas)
    - 60 minutes
- Refined starches (chips, crackers, bread)
    - 30 minutes
- Mixed meals (protein + fat + carb)
    - 20 minutes
- Protective snacks (cheese, nuts)
    - 0 minutes
- If you are xerostomic, 2x all times